

IT694 Computer Networks

**Dhirubhai Ambani Institute of Information and Communication Technology
(DA-IICT)**

First In-Semester Examination, Winter 2020

INSTRUCTIONS:

- Write your name and student number **clearly** at the top of each page. If you do not follow this you will get zero right away.
- There are 2 double sided sheets (Total 4 pages). Ensure that you have all the sheets.
- Your answers should be on the question paper in the space provided. Use the last side for calculations, be clear.
- You have **1.5 hours** to complete the test.
- No questions will be entertained by the invigilator.

1. A user needs to send the server some information. The request line method is
[1 point]
 - (i) GET
 - (ii) HEAD
 - (iii) SEND
 - (iv) POST
2. For which of the following applications select the appropriate transport protocol [2 points each]
 - (i) Streaming a live video over the internet: TCP / UDP
 - (ii) Instant messaging/email: TCP / UDP
 - (iii) Logging in to your bank website: TCP / UDP
 - (iv) Voice over IP: TCP / UDP
 - (v) Large file transfers: TCP / UDP
 - (vi) Looking up a very small value from a directory service: TCP / UDP
3. Suppose the algorithms used to implement the operations at layer k is changed. How does this impact operation at layers $k - 1$ and $k + 1$? [2 points]
4. Assume propagation delay is less than transmission delay on a link connecting host A to B. If host A starts transmission at $t=0$, at time equal to transmission delay, where is the first bit of the packet? [1 point]
 - (i) At Host A
 - (ii) In the link
 - (iii) At Host B
5. HTTP is ? [1 point]
 - (i) The language used to code web pages
 - (ii) A file transfer protocol
 - (iii) The protocol that is used for web pages, that transfers hypertext
 - (iv) The web browser
6. State True or False [2 points each]:
 - (i) When a client browser requests a web page and the page is sent back, a direct physical connection between client and server is formed while the information is being sent:
 - (ii) The Internet is based on the concept of circuit switching, in contrast to the tradition telephone network, which is based on the concept of packet switching:

- (iii) On a low-speed transmission link, the “width” of each bit will be greater than the “width” of a bit on a high-speed transmission link:
 - (iv) A host machine is a network node that runs a user application:
 - (v) Encapsulation is the process of capturing and reading data as it travels across the Internet:
7. A user wishes to send a message of size 1 MB (MegaByte) to another user over a point-to-point link. The message is divided into blocks of size 1000 bytes. Each of the 5 layers add a header of length 20 bytes each to the block as it goes down the protocol stack. What is the “user” throughput (as seen by the end-user) if the link capacity is 10 Mbps. Ignore processing and propagation delays. Express the answer in Mbps. [2 points]
8. Suppose a 128-Kbps point-to-point link is set up between Earth and a rover on Mars. The distance from Earth to Mars (when they are closest together) is approximately 55×10^9 m and data travels over the link at the speed of light 3×10^8 m/sec.
- (i) Calculate the minimum RTT for the link. [2 points]
 - (ii) Calculate the delay $\times$ bandwidth product for the link. [2 point]
 - (iii) A camera on the rover takes pictures of its surroundings and sends these to Earth. How quickly after a picture is taken can it reach Mission Control on Earth? Assume that each image is 5 MB in size [2 points]

NAME:

STUDENT NUMBER:

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